3 Connectivity

Our definition of k-connectedness, given in Chapter 1.4, is somewhat unintuitive. It does not tell us much about 'connections' in a k-connected graph: all it says is that we need at least k vertices to *dis*connect it. The following definition – which, incidentally, implies the one above – might have been more descriptive: 'a graph is k-*connected* if any two of its vertices can be joined by k independent paths'.

It is one of the classic results of graph theory that these two definitions are in fact equivalent, are dual aspects of the same property. We shall study this theorem of Menger (1927) in some depth in Section 3.3.

In Sections 3.1 and 3.2, we investigate the structure of the 2-connected and the 3-connected graphs. For these small values of k it is still possible to give a simple general description of how these graphs can be constructed.

In Sections 3.4 and 3.5 we look at other concepts of connectedness, more recent than the standard one but no less important: the number of H-paths in G for a subgraph H of G, and the existence of disjoint paths in G linking up specified pairs of vertices.

3.1 2-Connected graphs and subgraphs

The simplest 2-connected graphs are the cycles. All the others can be constructed inductively from a cycle by adding paths:

Proposition 3.1.1. *A graph is 2-connected if and only if it can be constructed from a cycle by successively adding H-paths to graphs H already constructed (Fig. 3.1.1).* [4.2.6]

Proof. Clearly, every graph constructed as described is 2-connected. Conversely, let a 2-connected graph G be given. Then G contains a

R. Diestel, *Graph Theory*, Graduate Texts in Mathematics 173,
https://doi.org/10.1007/978-3-662-70107-2_3

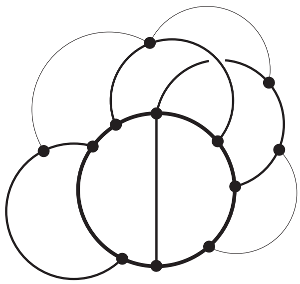

Fig. 3.1.1. The construction of 2-connected graphs

H cycle, and hence has a maximal subgraph H constructible as above. Since any edge $xy \in E(G) \setminus E(H)$ with $x, y \in H$ would define an H-path, H is an induced subgraph of G. Thus if $H \neq G$, then by the connectedness of G there is an edge vw with $v \in G - H$ and $w \in H$. As G is 2-connected, $G - w$ contains a v–H path P. Then wvP is an H-path in G, and $H \cup wvP$ is a constructible subgraph of G larger than H. This contradicts the maximality of H. □

Just as an arbitrary graph can be decomposed into its maximal connected subgraphs, or *components*, we can try to decompose a connected graph G into its maximal 2-connected subgraphs. These may not quite be disjoint, and they may not quite cover all of G. However, it is easy to weaken the notion of 'maximal 2-connected subgraph' slightly so that the subgraphs fitting the weaker notion do cover G and are still nearly disjoint. These 'blocks' fit together nicely in a tree-like fashion, which captures precisely the overall structure of G in terms of those blocks.

block Formally, a *block* is a maximal connected subgraph without a cutvertex.[1] Thus, every block is either a maximal 2-connected subgraph, or a bridge (with its ends), or an isolated vertex. Conversely, every such subgraph is a block. By their maximality, different blocks of G overlap in at most one vertex, which is then a cutvertex of G. Hence every edge of G lies in a unique block, and G is the union of its blocks.

Cycles and bonds are confined to a single block:

[4.6] **Lemma 3.1.2.** *Let G be any graph.*

(i) *The cycles in G are precisely the cycles in its blocks.*

(ii) *The bonds of G are precisely the bonds of its blocks.*

Proof. (i) Any cycle in G is a connected subgraph without a cutvertex, and hence lies in some maximal such subgraph. By definition, this is a block of G.

[1] . . . of the subgraph; it may contain cutvertices of G.

(ii) The proof follows easily by repeated application of the following observation. Consider any cut in G. Let xy be one of its edges, and B the block containing it. By the maximality of B in the definition of a block, G contains no B-path. Hence every x–y path in G lies in B, so those edges of our cut that lie in B separate x from y even in G. □

As every edge lies in a unique block, belonging to a common block is an equivalence relation on the edge set of a graph. This equivalence can be expressed in two other interesting ways:

Lemma 3.1.3. *The following statements are equivalent for distinct edges e, f of a graph G:* [4.6]

(i) *The edges e, f belong to a common block of G.*

(ii) *The edges e, f belong to a common cycle in G.*

(iii) *The edges e, f belong to a common bond of G.*

Proof. (i)→(ii) It clearly suffices to prove that in a 2-connected graph any two 2-sets of vertices can be joined by two disjoint paths. This follows easily by induction based on Proposition 3.1.1.[2]

(ii)→(iii) Deleting e and f from a cycle $C \ni e, f$ leaves a partition of $V(C)$ into two connected sets. Extend this to a partitition into two connected sets of the vertex set of the component of G containing C. (How?) The edges between these sets form a bond of G containing e and f.

(iii)→(i) By Lemma 3.1.2 (ii), two edges can lie in a common bond only if they belong to the same block. □

Our last lemma on blocks shows how they fit together to form the coarse structure of G. Let A denote the set of cutvertices of G, and $\mathcal{B}$ the set of its blocks. The bipartite graph on $A \cup \mathcal{B}$ formed by the edges aB with $a \in B$ is the *block graph* of G, see Figure 3.1.2. *block graph*

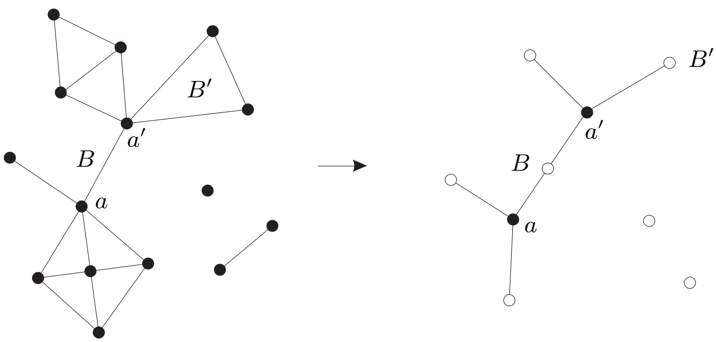

Fig. 3.1.2. A graph and its block graph

[2] See Exercise 5. Note that this is the case $k = 2$ of Menger's theorem (3.3.1).

Lemma 3.1.4. *The block graph of a connected graph is a tree.* □

Lemma 3.1.4 generalizes to graphs of higher connectivity: every $(k-1)$-connected graph has a canonical tree-like decomposition that separates all its 'k-blocks'. See Theorem 12.3.7 for the precise statement, and Exercise 17 in Chapter 12 for the case of $k = 3$.

3.2 The structure of 3-connected graphs

In this section we describe how every 3-connected graph can be obtained from a K^4 by a succession of elementary operations preserving 3-connectedness. We then prove a theorem of Tutte about the algebraic structure of the cycle space of 3-connected graphs; this will play an important role again in Chapter 4.5.

Proposition 3.1.1 describes how the 2-connected graphs can be constructed inductively, starting from a cycle. All the graphs constructed in the process were themselves 2-connected, so the graphs constructible in this way are precisely the 2-connected graphs. We shall now do something similar for 3-connected graphs. We shall prove that every 3-connected graph $G \neq K^4$ can be turned into a smaller 3-connected graph in two ways: by deleting an edge (and suppressing any vertices of degree 2 that may arise), and by contracting an edge. Inverting these processes will give us two independent ways of building all 3-connected graphs from a K^4.

$G \dot{-} e$ Given an edge e in a graph G, we write $G \dot{-} e$ for the *multigraph* obtained from $G - e$ by suppressing any end of e that has degree 2 in $G - e$.[3]

Lemma 3.2.1. *Let e be an edge in a graph G. If $G \dot{-} e$ is 3-connected, then so is G.*

(1.4.2) *Proof.* Thinking of G as obtained from $G \dot{-} e$ by adding e, let us call the vertices of $G \dot{-} e$ the *old* vertices of G, and any other vertex of G (which will be an end of e) a *new* vertex. Remembering that $G \dot{-} e$, being 3-connected, has no parallel edges, it is easy to see that, in G, no two vertices x_1, x_2 can separate a new vertex from all the old vertices. So it suffices to show that $\{x_1, x_2\}$ cannot separate two old vertices. If they did, then those old vertices would be separated in $G \dot{-} e$ by x'_1 and x'_2, where either $x'_i = x_i$ or, if x_i is new, x'_i is the edge of $G \dot{-} e$ subdivided by x_i. By Proposition 1.4.2, this contradicts our assumption that $G \dot{-} e$ is 3-connected. □

[3] See Chapter 1.10 for the formal definition of suppressing vertices in a multigraph. Recall also that 3-connected multigraphs cannot have multiple edges. Since parallel edges arising when a vertex is suppressed are not deleted, our assumption in Lemma 3.2.1 that the multigraph $G \dot{-} e$ is 3-connected implies that no parallel edges arise when it is formed from the graph G. Thus $G \dot{-} e$, too, is in fact a graph.